

|                           |   |                        |          |             |   |              |
|---------------------------|---|------------------------|----------|-------------|---|--------------|
| <b>Document reference</b> |   | NSP/003/010            |          |             |   |              |
| <b>Version:-</b>          | 2 | <b>Date of Issue:-</b> | Nov - 08 | <b>Page</b> | 1 | <b>of</b> 13 |

# NPS/003/010 - Technical Specification for 400V LVAC Boards utilised in Major Substations

## 1. Purpose

This document is the technical specification of 400V LVAC boards for use in major substations on CE Electric UK's distribution network.

This document supersedes the following documents, all copies of which should be destroyed.

| Ref         | Version | Title                                                                      |
|-------------|---------|----------------------------------------------------------------------------|
| NPS/003/010 | 1       | Technical Specification for 400V LVAC Boards utilised in Major Substations |

## 2. Scope

This specification covers the technical requirements for 400V AC switchgear, control gear and fusegear (LVAC boards) for use in supply points and primary substations on CE Electric UK's distribution network.

Any project specific requirements as detailed in Appendix 3 - Addendum to Supplier Requirements.

## 3. Technical Requirements

### General

Switchboards shall be 3Ø modular construction complying with BS EN 60439-1, minimum classification Form 4 Type 2. They shall be suitable for terminating PVC insulated and sheathed multicore, armoured cables in a range of sizes to match the rating of the associated circuit (max 70mm<sup>2</sup> on incomers) using brass compression type glands.

### Indoor Switchboards

Switchboard shall typically comprise two incoming and seven outgoing ways, with ratings as per the attached Appendices . The board will have an integral:

3Ø 12w MCB protected distribution board for lighting, sockets etc  
& a 2kVA 230/110v transformer / 4w DP MCB protected distribution board for 110v supplies

(MCB's shall be rated and tested as a point of isolation and shall normally be type B or D - detailed requirements are listed in the Appendices)

Each incoming circuit will normally supply its respective tap change and cooler supply, with one of the circuit s supplying the remainder of the board. Auto changeover facilities will be provided to ensure supplies are maintained to all outgoing circuits if the normal supply becomes unavailable.

Three transformer substation sites will have an additional incoming and outgoing way.

|                           |   |                        |          |             |   |              |
|---------------------------|---|------------------------|----------|-------------|---|--------------|
| <b>Document reference</b> |   | NSP/003/010            |          |             |   |              |
| <b>Version:-</b>          | 2 | <b>Date of Issue:-</b> | Nov - 08 | <b>Page</b> | 2 | <b>of</b> 13 |

## Outdoor Switchboards

Switchboard comprise an incoming 3 position switch (on / off / on) and five outgoing ways, with ratings as per the attached Appendices. One of the ways will control an integral 3Ø outdoor socket, with another supplying an indoor switchboard.

Appendix 1 of this document details the options selected within BS EN 60439-1 and clarifies, amends or supplements BS EN 60439-1 to meet the requirements of CE Electric UK.

Appendix 2 is a Declaration of Technical Performance to be completed for each switchboard.

Appendix 3 contains an Addendum to Supplier Requirements which lists project specific requirements

## 4. References

### 4.1 External Documentation

| Reference     | Title                                                                                                        |
|---------------|--------------------------------------------------------------------------------------------------------------|
| BS EN 60439-1 | Low Voltage Switchgear and Control Gear Assemblies: part 1: Type Tested and Partially Type Tested Assemblies |
| BS EN 60529   | Specification for Degree of Protection provided by Enclosure                                                 |
| BS EN 60898   | Circuit Breakers for Over Current Protection for Households and similar Installations                        |

### 4.2 Internal documentation

| Reference | Title |
|-----------|-------|
|-----------|-------|

## 5 Definitions

| Term | Definition |
|------|------------|
|------|------------|

|                           |   |                        |          |             |   |           |    |
|---------------------------|---|------------------------|----------|-------------|---|-----------|----|
| <b>Document reference</b> |   | NSP/003/010            |          |             |   |           |    |
| <b>Version:-</b>          | 2 | <b>Date of Issue:-</b> | Nov - 08 | <b>Page</b> | 3 | <b>of</b> | 13 |

## 6. Authority for issue

### 6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

|         |                    |         |          |
|---------|--------------------|---------|----------|
|         |                    | Sign    | Date     |
| V Gorny | Standards Engineer | V Gorny | 27/10/08 |

### 6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

|             |                 |             |          |
|-------------|-----------------|-------------|----------|
|             |                 | Sign        | Date     |
| M Nicholson | System Strategy | M Nicholson | 30/10/08 |

### 6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

|              |                    |              |          |
|--------------|--------------------|--------------|----------|
|              |                    | Sign         | Date     |
| D Blackledge | Standards Engineer | D Blackledge | 27/10/08 |

### 6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

|           |                   |           |          |
|-----------|-------------------|-----------|----------|
|           |                   | Sign      | Date     |
| S.Johnson | DBD Administrator | S.Johnson | 21/11/08 |

### 6.5 Approval

Approval is given for the content of this document

|              |                                |              |          |
|--------------|--------------------------------|--------------|----------|
|              |                                | Sign         | Date     |
| C.Holdsworth | Standards & Assessment Manager | C.Holdsworth | 10/11/08 |

### 6.6 Authorisation

Authorisation is granted for publication of this document

|          |                     |          |          |
|----------|---------------------|----------|----------|
|          |                     | Sign     | Date     |
| C.Morgan | Head of Procurement | C.Morgan | 10/11/08 |

|                           |             |                        |          |             |         |
|---------------------------|-------------|------------------------|----------|-------------|---------|
| <b>Document reference</b> | NSP/003/010 |                        |          |             |         |
| <b>Version:-</b>          | 2           | <b>Date of Issue:-</b> | Nov - 08 | <b>Page</b> | 4 of 13 |

## Appendix 1: Technical Specification Sheet

| Substation LVAC Board Specification                                 |      |                                                                                                                                                    |  |
|---------------------------------------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------|--|
|                                                                     | Unit | Requirement                                                                                                                                        |  |
| Supply Voltage                                                      | V    | 400 +10% -6%                                                                                                                                       |  |
| No of Phases                                                        |      | 3                                                                                                                                                  |  |
| Frequency                                                           | Hz   | 50                                                                                                                                                 |  |
| Internal Separation                                                 |      | Form 4 Type 2                                                                                                                                      |  |
| Normal Current Rating:<br>Incomer Fuse Switches or<br>ACBs.         |      | As specified in Appendix 3                                                                                                                         |  |
| Normal Current Rating: Fuse<br>Switches or ACBs.                    |      | As specified in Appendix 3                                                                                                                         |  |
| Normal Current Rating:<br>Incomer automatic changeover<br>contactor | A    | 100A or 200A (As specified - see Appendix 3)                                                                                                       |  |
| Busbar Normal Current Rating                                        | A    | 100A or 200A (As specified - seen Appendix 3)                                                                                                      |  |
| Fuse Switch Ratings                                                 |      | As specified in Appendix 3                                                                                                                         |  |
| Fuse Ratings                                                        |      | As specified in Appendix 3                                                                                                                         |  |
| Fuse Specification                                                  |      | BS 88 Parts 1 and 2                                                                                                                                |  |
| Miniature Circuit Breaker<br>Specification                          |      | BS EN 60898                                                                                                                                        |  |
| Prospective current                                                 | kA   | 35                                                                                                                                                 |  |
| Fault Current Withstand<br>Rating.                                  | kA   | 35                                                                                                                                                 |  |
| Fault Current Duration                                              | s    | 1                                                                                                                                                  |  |
| Incoming Supply Cable Entry                                         |      | Bottom                                                                                                                                             |  |
| Outgoing Cables Exit                                                |      | Bottom                                                                                                                                             |  |
| Gland plate: Min height above<br>ground level.                      | mm   | 450                                                                                                                                                |  |
| Space below gland plate                                             |      | Must be enclosed and vermin proof                                                                                                                  |  |
| Access: Installation                                                |      | Front                                                                                                                                              |  |
| Access: Operation                                                   |      | Front                                                                                                                                              |  |
| Access: Maintenance                                                 |      | Front                                                                                                                                              |  |
| Switchboard Position                                                |      | Capable of being mounted against a wall.                                                                                                           |  |
| Locking Arrangements                                                |      | Isolators / Switch fuses shall be suitable for locking-off with padlocks with a 5mm diameter shackle.                                              |  |
|                                                                     |      |                                                                                                                                                    |  |
| Minimum Degree Protection of<br>Enclosures (IEC60529)               |      | Indoor Boards: IP41D<br>Outdoor Boards: IP53D                                                                                                      |  |
| Door stays                                                          |      | All access must be fitted with door stays                                                                                                          |  |
| Testing lamp facility                                               |      | Incoming terminals should access to for testing without the need to remove covers (suitable for use typically with Drummond or Cyclim test lamps). |  |
| EMC regulations                                                     |      | Boards shall be EMC compatible and CE marked                                                                                                       |  |
| Heating                                                             |      | Outdoor switchboards only: A fused thermostatically controlled 110V heater shall be installed.                                                     |  |
| Lighting                                                            |      | Outdoor switchboards only: A fused 110V light shall be installed.                                                                                  |  |

|                           |   |                        |        |             |   |           |    |
|---------------------------|---|------------------------|--------|-------------|---|-----------|----|
| <b>Document reference</b> |   | NPS/003/010            |        |             |   |           |    |
| <b>Version:-</b>          | 2 | <b>Date of Issue:-</b> | Oct-08 | <b>Page</b> | 5 | <b>of</b> | 13 |

## Appendix 2 - SELF CERTIFICATION CONFORMANCE DECLARATION

LVAC Boards are required to be supplied required to be supplied against this specification shall comply with the latest issues of the relevant ENATS, British and International Standards specified. The following tables are intended to amplify and/or clarify the requirements of elements of these Standards but do not preclude meeting all requirements of the standards. .

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

### Instructions for completion

- When Cs1 code is entered no remark is necessary.
- When any other code is entered the reason for non-conformance shall be entered.
- Prefix each remark with the relevant 'BS EN' 'IEC' or 'ENATS' as appropriate.

**Manufacturer:**

**Product Reference:**

**Name:**

**Signature:**

**Date:**

**NOTE:**

|                           |             |                        |        |             |   |              |
|---------------------------|-------------|------------------------|--------|-------------|---|--------------|
| <b>Document reference</b> | NPS/003/010 |                        |        |             |   |              |
| <b>Version:-</b>          | 2           | <b>Date of Issue:-</b> | Oct-08 | <b>Page</b> | 6 | <b>of</b> 13 |

| <b>Substation LVAC Board Specification</b> |                                          |                                                                                                 |                  |                    |
|--------------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------|------------------|--------------------|
|                                            |                                          | Clause / Requirements                                                                           | Conformance Code | Remarks / Comments |
|                                            | Supply Voltage                           | 400v +10% -6%                                                                                   |                  |                    |
|                                            | No of Phases                             | 3                                                                                               |                  |                    |
|                                            | Frequency                                | 50 Hz                                                                                           |                  |                    |
|                                            | Internal Separation                      | Form 4 Type 2                                                                                   |                  |                    |
|                                            | Incomers: Fuse Switch / ACB              | 100A ) Depending on LVAC Board Type<br>200A ) as Specified in Appendix 3                        |                  |                    |
|                                            | Outgoing Circuits: Fuse Switch / ACB     | 100A ) Depending on<br>200A ) LVAC Board Type<br>63A ) Type as Specified<br>32A ) in Appendix 3 |                  |                    |
|                                            | Busbar Normal Current Rating             | 100A ) Depending on LVAC Board Type<br>200A ) as Specified in Appendix 3                        |                  |                    |
|                                            | Fuses                                    | BS EN 88 Parts 1 and 2                                                                          |                  |                    |
|                                            | MCB's                                    | BS EN 60898                                                                                     |                  |                    |
|                                            |                                          | Suitable for Point of Isolation & Lockable                                                      |                  |                    |
|                                            | Fault break duty or prospective current? | 35kA                                                                                            |                  |                    |
|                                            | Rated Short Time Withstand Current.      | 35kA                                                                                            |                  |                    |
|                                            | Rated Time Duration                      | 1 Sec                                                                                           |                  |                    |
|                                            | Incomer Cable                            | Bottom Entry                                                                                    |                  |                    |
|                                            | Feeder Cable                             | Bottom Entry                                                                                    |                  |                    |
|                                            | Gland plate                              | Min 450mm above Ground / Floor level                                                            |                  |                    |
|                                            | Space below the gland plate              | Enclosed and vermin proof. State method of enclosure.                                           |                  |                    |
|                                            | Installation                             | Suitable for mounting against wall                                                              |                  |                    |
|                                            | Operation                                | From Front                                                                                      |                  |                    |
|                                            | Maintenance                              | From Front                                                                                      |                  |                    |

|                           |   |                        |        |             |   |           |    |
|---------------------------|---|------------------------|--------|-------------|---|-----------|----|
| <b>Document reference</b> |   | NPS/003/010            |        |             |   |           |    |
| <b>Version:-</b>          | 2 | <b>Date of Issue:-</b> | Oct-08 | <b>Page</b> | 7 | <b>of</b> | 13 |

| <b>Substation LVAC Board Specification</b> |                                                         |                                                                                                                     |                  |                    |
|--------------------------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------|--------------------|
|                                            |                                                         | Clause / Requirements                                                                                               | Conformance Code | Remarks / Comments |
|                                            | Incoming and outgoing isolators / switch fuses          | Lockable with a padlock with a 5mm diameter shackle                                                                 |                  |                    |
|                                            | Minimum Degree Protection of Enclosures: Indoor Boards. | Indoor Boards: IP41D                                                                                                |                  |                    |
|                                            | Door stays                                              | Required                                                                                                            |                  |                    |
|                                            | Test lamp facilities                                    | Testing access to incoming terminals without the need to remove covers – (typically Drummond or Cyclim test lamps). |                  |                    |
|                                            | EMC requirements                                        | EMC compatible with requirements of BS EN 60439 and CE marked                                                       |                  |                    |
|                                            | Outdoor switchboards                                    | Fused 110V light Installed.                                                                                         |                  |                    |
|                                            |                                                         | Fused thermostatically controlled 110V heater installed.                                                            |                  |                    |
|                                            |                                                         | Min Degree of Protection : IP53D                                                                                    |                  |                    |

|                           |   |                        |        |             |   |           |    |
|---------------------------|---|------------------------|--------|-------------|---|-----------|----|
| <b>Document reference</b> |   | NPS/003/010            |        |             |   |           |    |
| <b>Version:-</b>          | 2 | <b>Date of Issue:-</b> | Oct-08 | <b>Page</b> | 8 | <b>of</b> | 13 |

---

## Appendix 3: Addendum of Supplier Requirements

Typical arrangements – however these may need amended for specific project based applications by Primary Engineering Projects.

- 1.0 The switchgear and control requirements are detailed in the following tables:

|                           |             |                        |        |             |         |
|---------------------------|-------------|------------------------|--------|-------------|---------|
| <b>Document reference</b> | NPS/003/010 |                        |        |             |         |
| <b>Version:-</b>          | 2           | <b>Date of Issue:-</b> | Oct-08 | <b>Page</b> | 9 of 13 |

## 1.1 PRIMARY / GRID SUBSTATION **INDOOR** LVAC BOARD (1 or 2 Transformers - see associated drawing)

| <u>Way Designation</u><br><i>(label reqd in bold)</i>                                                                    | <u>Type</u>                                                                   | <u>Switch Rating</u> | <u>Fuse Rating</u> | <u>Other</u>                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Incomer 1</b><br><b>Incomer 2</b>                                                                                     | TP+N<br>Switch fuse                                                           | 100A                 | 100A               | Glanding / termination of incoming cable to be in a separate cubicle on the switchboard with shrouded links. Access required to allow phase checks between incomers.<br>Suitable for cable cores up to 70mm <sup>2</sup> .<br><b>NOTE 1:</b> Each incomer will normally supply its respective transformer tap change & cooler supply – but will auto changeover to alternative if supply is interrupted. |
| <b>Auto Changeover Arrangement</b>                                                                                       |                                                                               | 100A                 |                    | Changeover contactor(s) required between incomers. On loss of supply, automatic changeover to alternative incomer. (Normally biased to incomer 1). Should incorporate indicator lights showing status incoming circuits / changeover contactor.                                                                                                                                                          |
| <b>Test Way</b>                                                                                                          | TP+N<br>Switch fuse                                                           | 100A                 | 100A               | To be used for temporary single core flexible cables. A flap should be included on the switch fuse door to allow egress of the flexible cables.<br><b>NOTE 2:</b> Can be used to make external generator connection to supply board when both incoming supplies unavailable                                                                                                                              |
| <b>T1 Tap Change &amp; Cooler Supply</b>                                                                                 | TP+N<br>Switch fuse                                                           | 63A                  | 63A                | <b>See NOTE 1</b> above                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>T2 Tap Changer &amp; Cooler Supply</b>                                                                                | TP+N<br>Switch fuse                                                           | 63A                  | 63A                | <b>See NOTE 1</b> above                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Distribution below normally supplied via Incomer 1 - will auto changeover to alternative if incomer not available</b> |                                                                               |                      |                    |                                                                                                                                                                                                                                                                                                                                                                                                          |
| Fuse board 1 –<br><b>DB/11kV SP + LTG</b>                                                                                | TP+N<br>Switch<br>disconnecter<br>incomer<br>3 Ø x 12<br>way:<br>(split 8 +4) | 63A                  | 63A                | 3 Ø x 12 way (36 x single pole lockable devices)<br><br>Heating circuits to be contactor controlled by externally mounted thermostat and 1 hour override timer mounted on the LVAC board.<br>Terminals required to accept externally mounted thermostat wiring.<br><br>(See device type, rating & labelling requirements)                                                                                |
| Supply to<br>230/110V, 2kVA<br>Transformer –<br><b>DB/110v SP + N</b>                                                    | SP+N<br>Switch fuse                                                           | 32A                  | 32A                | 2kVA 230/110V centre earth tapped transformer.<br>Double-pole switch disconnecter to provide 110V isolation<br>Fuseboard 2 -110V: 4 x 6A lockable DP MCBs                                                                                                                                                                                                                                                |
| 2 x Spare –<br><b>Spare DB/33kV</b>                                                                                      | TP+N<br>Switch fuse                                                           | 63A                  | 63A                |                                                                                                                                                                                                                                                                                                                                                                                                          |

|                           |             |                        |        |             |          |
|---------------------------|-------------|------------------------|--------|-------------|----------|
| <b>Document reference</b> | NPS/003/010 |                        |        |             |          |
| <b>Version:-</b>          | 2           | <b>Date of Issue:-</b> | Oct-08 | <b>Page</b> | 10 of 13 |

## 1.2 GRID SUBSTATION **INDOOR** LVAC BOARD (3 Transformer Sites - see associated drawing)

| <u>Way Designation</u><br><i>(label reqd in bold)</i>                                                                    | <u>Type</u>                                                           | <u>Switch Rating</u> | <u>Fuse Rating</u> | <u>Other</u>                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Incomer 1</b><br><b>Incomer 2</b><br><b>Incomer 3</b>                                                                 | TP+N<br>Switch fuse                                                   | 100A                 | 100A               | Glanding / termination of incoming cable to be in a separate cubicle on the switchboard with shrouded links. Access required to allow phase checks between incomers.<br>Suitable for cable cores up to 70mm <sup>2</sup> .<br><b>NOTE 1:</b> Each incomer will normally supply its respective transformer tap change & cooler supply – but will auto changeover to alternative if supply is interrupted. |
| <b>Auto Changeover Arrangement</b>                                                                                       |                                                                       | 100A                 |                    | Changeover contactor(s) required between incomers. On loss of supply, automatic changeover to alternative incomer. (Normally biased to incomer 1). Should incorporate indicator lights showing status incoming circuits / changeover contactor.                                                                                                                                                          |
| <b>Test Way</b>                                                                                                          | TP+N<br>Switch fuse                                                   | 100A                 | 100A               | To be used for temporary single core flexible cables. A flap should be included on the switch fuse door to allow egress of the flexible cables.<br><b>NOTE 2:</b> Can be used to make external generator connection to supply board when both incoming supplies unavailable                                                                                                                              |
| <b>T1 Tap Change &amp; Cooler Supply</b>                                                                                 | TP+N<br>Switch fuse                                                   | 63A                  | 63A                | <b>See NOTE 1</b> above                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>T2 Tap Changer &amp; Cooler Supply</b>                                                                                | TP+N<br>Switch fuse                                                   | 63A                  | 63A                | <b>See NOTE 1</b> above                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>T3 Tap Changer &amp; Cooler Supply</b>                                                                                | TP+N<br>Switch fuse                                                   | 63A                  | 63A                | <b>See NOTE 1</b> above                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Distribution below normally supplied via Incomer 1 - will auto changeover to alternative if incomer not available</b> |                                                                       |                      |                    |                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Fuse board 1 – DB/11kV SP + LTG</b>                                                                                   | TP+N<br>Switch disconnecter incomer<br>3 Ø x 12 way :<br>(split 8 +4) | 63A                  | 63A                | 3 Ø x 12 way (36 x single pole lockable devices)<br><br>Heating circuits to be contactor controlled by externally mounted thermostat and 1 hour override timer mounted on the LVAC board.<br>Terminals required to accept externally mounted thermostat wiring.<br>(See device type, rating & labelling requirements)                                                                                    |
| <b>Supply to 230/110V, 2kVA Transformer – DB/110v SP + N</b>                                                             | SP+N<br>Switch fuse                                                   | 32A                  | 32A                | 2kVA 230/110V centre earth tapped transformer.<br>Double-pole switch disconnecter to provide 110V isolation<br>Fuseboard 2 -110V: 4 x 6A lockable DP MCBs                                                                                                                                                                                                                                                |
| <b>2 x Spare – Spare DB/33kV</b>                                                                                         | TP+N<br>Switch fuse                                                   | 63A                  | 63A                |                                                                                                                                                                                                                                                                                                                                                                                                          |

|                           |   |                        |        |             |    |              |
|---------------------------|---|------------------------|--------|-------------|----|--------------|
| <b>Document reference</b> |   | NPS/003/010            |        |             |    |              |
| <b>Version:-</b>          | 2 | <b>Date of Issue:-</b> | Oct-08 | <b>Page</b> | 11 | <b>of</b> 13 |

## 1.3 GRID SUBSTATION **OUTDOOR** LVAC BOARD (2 or 3 Transformer Sites - see associated drawing)

| <u>Way Designation</u><br><i>(label reqd in bold)</i>                                                            | <u>Type</u>                                                                                         | <u>Switch Rating</u> | <u>Fuse Rating</u> | <u>Other</u>                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Incoming Supply From T*Aux Transformer</b><br><br><b>Off</b><br><br><b>Incoming supply from T* LVAC Board</b> | 3 position manual changeover switch handle with indications and mechanical interlocking on incomers | 200A                 | 200A               | Should have a centre OFF position and be pad lockable in all positions.<br><br>Solid links in place of fuses for manual changeover device. Glanding / termination of incoming cable to be in a separate cubicle on the switchboard with shrouded links. Access required to allow phase checks between incomers. Should be suitable for cable cores up to 70mm <sup>2</sup> . |
| <b>Outgoing supply to T* LVAC board</b>                                                                          | TP&N switch fuse                                                                                    | 200A                 | 200A               |                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Outgoing to Indoor LVAC Switchboard</b>                                                                       | TP&N switch fuse                                                                                    | 100A                 | 100A               |                                                                                                                                                                                                                                                                                                                                                                              |
| <b>T* Tap Change &amp; Cooler Supply</b>                                                                         | TP&N switch fuse                                                                                    | 63A                  | 63A                |                                                                                                                                                                                                                                                                                                                                                                              |
| <b>3 Phase Socket</b>                                                                                            | TP&N switch fuse and 63A 3ph socket                                                                 | 63A                  | 63A                |                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Spare</b>                                                                                                     | TP&N switch fuse                                                                                    | 63A                  | 63A                |                                                                                                                                                                                                                                                                                                                                                                              |

\* To be specified at time of ordering

|                           |             |                        |        |             |          |
|---------------------------|-------------|------------------------|--------|-------------|----------|
| <b>Document reference</b> | NPS/003/010 |                        |        |             |          |
| <b>Version:-</b>          | 2           | <b>Date of Issue:-</b> | Oct-08 | <b>Page</b> | 12 of 13 |

## 1.4 LVAC BOARD - DB/11kV SP + LTG Arrangement

| DB/11kV SP + LTG Arrangement and Labelling Requirement |        |            |                                             |          |
|--------------------------------------------------------|--------|------------|---------------------------------------------|----------|
| Way                                                    | Device | Rating (A) | Description                                 | Comments |
| 1L1                                                    | MCB    | 32         | Spare                                       |          |
| 2L1                                                    | MCB    | 32         | 110v DC Battery & Charger                   | TYPE C   |
| 3L1                                                    | MCB    | 16         | Spare                                       |          |
| 2L1                                                    | RCD    | 32         | Sockets Control Room                        |          |
| 2L2                                                    | RCD    | 32         | Sockets Switch Room                         |          |
| 2L3                                                    | MCB    | 16         | 24v DC Battery & Charger                    | TYPE C   |
| 3L1                                                    | MCB    | 16         | Lighting : Control & Switch Rooms. WC & MTG |          |
| 3L2                                                    | MCB    | 6          | Lighting : External Switch Room Lighting    |          |
| 3L3                                                    | MCB    | 6          | Lighting : Control Room                     |          |
| 4L1                                                    | MCB    | 6          | Intruder Alarm                              |          |
| 4L2                                                    | MCB    | 6          | Fire alarm                                  |          |
| 4L3                                                    | MCB    | 6          | Spare                                       |          |
| 5L1                                                    | MCB    | 16         | Small Power : Heating Metering Annexe       | TYPE C   |
| 5L2                                                    | MCB    | 16         | Small Power : Hand Dryer WC                 | TYPE C   |
| 5L3                                                    | MCB    | 16         | Small Power : Water Heater Toilet           |          |
| 6L1                                                    | MCB    | 40         | DB TR EN 1                                  | TYPE D   |
| 6L2                                                    | MCB    | 6          | Tx 1 Bund Pump Control                      |          |
| 6L3                                                    | MCB    | 16         | Spare                                       |          |
| 7L1                                                    | MCB    | 40         | DB TR EN 2                                  | TYPE D   |
| 7L2                                                    | MCB    | 16         | Tx 2 Bund Pump Control                      |          |
| 7L3                                                    | MCB    | 6          | Spare                                       |          |
| 8L1                                                    | MCB    | 32         | 2kA 230/110v Transformer                    | TYPE D   |
| 8L2                                                    | MCB    | 16         | Spare                                       |          |
| 8L3                                                    | MCB    | 6          | Spare                                       |          |
| Heating Distribution (contactor controlled)            |        |            |                                             |          |
| 9L1                                                    |        |            | Spare                                       |          |
| 9L2                                                    |        |            | Spare                                       |          |
| 9L3                                                    |        |            | Spare                                       |          |
| 10L1                                                   | MCB    | 32         | Heating Switch Room                         |          |
| 10L2                                                   | MCB    | 32         | Heating Switch Room                         |          |
| 10L3                                                   |        |            | Spare                                       |          |
| 11L1                                                   |        |            | Spare                                       |          |
| 11L2                                                   |        |            | Spare                                       |          |
| 11L3                                                   | MCB    | 32         | Heating Control Room                        |          |
| 12L1                                                   | MCB    | 16         | Heating WC                                  |          |
| 12L2                                                   | MCB    | 16         | Heating Store Room                          |          |
| 12L3                                                   |        |            | Spare                                       |          |

|                           |   |                        |        |             |    |           |    |
|---------------------------|---|------------------------|--------|-------------|----|-----------|----|
| <b>Document reference</b> |   | NPS/003/010            |        |             |    |           |    |
| <b>Version:-</b>          | 2 | <b>Date of Issue:-</b> | Oct-08 | <b>Page</b> | 13 | <b>of</b> | 13 |

## 2.0 Mounting

Outdoor LVAC boards shall be suitable for mounting across two parallel flange channels (PFC) see YEDL / Freedom drawing 01-0706-47. The LVAC also must be suitable to mount over a concrete cable trench for bottom cable entry.

## 3.0 The following drawings are attached showing typical arrangements for the circuits on the switchboard

|                                                                          |
|--------------------------------------------------------------------------|
| PRIMARY / GRID SUBSTATION <b>INDOOR</b> LVAC BOARD (1 or 2 Transformers) |
| GRID SUBSTATION <b>INDOOR</b> LVAC BOARD (3 Transformers)                |
| GRID SUBSTATION <b>OUTDOOR</b> LVAC BOARD (2 or 3 Transformers)          |

## 4.0 The following drawings/information should be supplied:-

- A general arrangement drawing detailing the depth, height and length and the position of the main components of the switchboard.
- A single line diagram of the switchboard detailing the power and control circuits.
- A drawing showing the detail fixing arrangements.
- Full technical details of the main components of the switchboard; to include manufacturer and load/fault ratings.
- Proposals for the off-loading of the switchboard on site. It should be noted that risk assessments and method statements will be required prior to delivery and that if a crane delivery is proposed, trained operators and slingers will be required who should provide proof of competence (CITB certificates etc).

## 5.0 Delivery

Deliveries shall be made to substation sites within the CE Electric area specified at the time of ordering.